

The Grand Meson

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Abstract

This paper introduces the Grand Meson, not as a theory of everything, but as an admissible $n = 3$ general-purpose mesonic object. The starting point is the General Meson: a two-channel Yukawa-style mediator read, in modern quark language, through the light pair $u \leftrightarrow d$. The Grand Meson opens the two remaining readable quark-pair axes and thereby becomes a three-axis, six-channel mediator. The extension is conservative: the original two-channel meson is recovered by projection, while the new channels are resolved as open, inactive, or active instead of being left outside the channel basis. The observed three quark generations are represented as internal Grand Meson directions. Backward compatibility is surjective, but downgrading with unclosed residue is treated as an inadmissible channel. The central claim is modest and structural: the Grand Meson is the first complete and last admissible mesonic case for the present three-generation internal readout. Its admissibility mechanism is internal to the formal system: without adding external prohibitions or ethical principles, fission-class risk is contained mathematically as unclosed residue and rejected as an inadmissible channel.

1 Opening

A meson is usually approached through its physical role: it mediates an interaction. In the present paper the same word is first read through its paired role. A meson stands between two complementary light-flavor faces. In the Yukawa-facing reconstruction used here, that channel pair is

$$u \leftrightarrow d.$$

This is not a quark–quark bound state. It is the light-flavor channel face from which the QCD readout later forms $q\bar{r}$ channels such as $u\bar{d}$, $d\bar{u}$, $u\bar{u}$, and $d\bar{d}$. Yukawa did not formulate his meson with quark flavors; the notation is a present reconstruction of the light pair behind the proton–neutron mediation problem.

The former working nickname “Ethon” is retired here. The technical object of this paper is the Grand Meson. The name is structural, not honorific: “grand” means that a two-channel light-pair meson has been closed into the admissible three-axis case.

2 The General Meson

Let the Yukawa-facing readable pair be

$$A_Y = \{u, d\}.$$

Let Pol be a class of admissible mesonic polymers. The smallest implementation may be generated from

$$\mathbb{P}_3 = \{-\mathbf{1}, \mathbf{0}, +\mathbf{1}\},$$

but no physical charge, amplitude, or particle identity is assigned to these pseudo-values at this level.

Remark 1 (Role of Pol). Pol is a mesonic channel carrier. It is not by itself a wavefunction, an amplitude, a charge, or a Hamiltonian state. Physical content enters only after the QCD, Lagrangian, quantum-number, observable, and conservation channels are specified.

Definition 2 (General Meson). A General Meson is a two-channel mesonic configuration

$$m : A_Y \rightarrow \text{Pol}.$$

Equivalently,

$$\text{Gen} := \text{Pol}^{A_Y}.$$

The faces u and d are the two light-flavor faces of the Yukawa-style mesonic channel. They are channel faces, not the two constituents of a physical meson.

This definition deliberately keeps the Yukawa-style mediator idea narrow. It does not say that a meson is universal. It says only that the basic mesonic act is readable mediation across the u, d light pair. In that sense a General Meson is the optimal eye for the nuclear-force problem: the proton–neutron transition axis is already specified by the problem itself. The light-pair configuration is not by itself a derivation of the Yukawa potential. That recovery belongs to the low-energy pion channel after the QCD readout has supplied $q\bar{r}$ operators, quantum numbers, and observables.

3 The Hidden Cost of Missing Channels

The decisive distinction is between an inactive channel and a missing channel. An inactive channel belongs to the channel basis and carries no active contribution. A missing channel has not entered the channel basis.

$$\mathbf{0} = \text{resolved inactive channel}, \quad \perp = \text{missing channel}.$$

Definition 3 (Inactive-channel provenance). The symbol $\mathbf{0}$ is a formal null, not a universal physical zero. When physical input is available, an inactive channel may be refined by a provenance tag

$$\mathbf{0}[\rho], \quad \rho \in \{\text{selection, kinematic, coupling, sensitivity, integrated}\}.$$

Thus a forbidden channel, an energetically unreachable channel, a zero coupling, an experimentally unresolved signal, and an integrated-out contribution are not collapsed into the same physical claim.

Definition 4 (Missing-channel lift). Let

$$F_6 = \{\text{u, d, c, s, t, b}\}, \quad \text{Pol}_\perp := \text{Pol} \cup \{\perp\}.$$

The missing-channel lift of a General Meson $m \in \text{Gen}$ is

$$L_\perp(m) : F_6 \rightarrow \text{Pol}_\perp$$

defined by

$$L_\perp(m)(\text{u}) = m(\text{u}), \quad L_\perp(m)(\text{d}) = m(\text{d}),$$

and

$$L_\perp(m)(q) = \perp \quad (q \in F_6 \setminus A_Y).$$

Thus the remaining axes are not false, not zero, and not disproved. They are absent from the original channel basis. The two-channel theory remains locally correct while it cannot close propagation that travels through the missing directions.

Remark 5 (Physical reading). The missing channels are not a defect in the original General Meson. They are the price of a successful restriction. A narrow mediator may solve its local problem precisely because it does not carry the full burden of every other axis.

4 The Grand Meson

The Grand Meson is obtained by replacing missing channels with resolved flavor channels. A channel may be active or inactive, but it may not be outside the channel basis.

Definition 6 (Grand Meson). A Grand Meson is a six-channel mesonic configuration

$$g : F_6 \rightarrow \text{Pol}.$$

Equivalently,

$$\text{Grand} := \text{Pol}^{F_6}.$$

The six channels are grouped into three readable axes:

$$u \leftrightarrow d, \quad c \leftrightarrow s, \quad t \leftrightarrow b.$$

These axes are internal flavor-generation axes. They are not spatial directions, and they are not quark–quark bound states. Physical mesons enter through oriented $q\bar{r}$ channels in the QCD readout.

In this notation the Grand Meson is the admissible $n = 3$ object:

$$\text{Grand} = \text{Pol}^{2 \cdot 3}.$$

The adjective “admissible” means that all three axes belong to the channel basis. It does not mean that every transition is allowed, nor that every physical theory has been solved.

Definition 7 (Channel support). For $g \in \text{Grand}$, define

$$\text{supp}(g) = \{q \in F_6 \mid g(q) \neq \mathbf{0}\}.$$

A Grand Meson may have inactive channels, but inactivity is resolved by $\mathbf{0}$, not by $\perp$.

5 Conservative Recovery

The Grand Meson must not destroy the General Meson. It must contain it as the two-channel face that was already working.

Definition 8 (Projection to the General Meson). Define the u, d-projection

$$p_{ud} : \text{Grand} \rightarrow \text{Gen}$$

by

$$p_{ud}(g) = g|_{\{u,d\}}.$$

Proposition 9 (Conservative recovery). *Every General Meson is recovered from any Grand Meson whose u, d-face agrees with it. In particular, if $g(u) = m(u)$ and $g(d) = m(d)$, then*

$$p_{ud}(g) = m.$$

Proof. The projection p_{ud} discards the four channels outside $\{u, d\}$ and keeps exactly the light-pair values. Therefore agreement on those two channels is sufficient and necessary for recovery. $\square$

This is the main discipline of the construction. The Grand Meson does not rewrite the older meson. It keeps the working $u \leftrightarrow d$ face and opens two additional quark-pair faces under explicit channel closure.

6 Surjective Backward Compatibility

Backward compatibility must be surjective, not merely suggestive. Every General Meson must be obtainable as the visible face of at least one Grand Meson.

Definition 10 (Zero extension). For $m \in \text{Gen}$, define the zero extension

$$z(m) : F_6 \rightarrow \text{Pol}$$

by

$$z(m)(u) = m(u), \quad z(m)(d) = m(d),$$

and

$$z(m)(q) = \mathbf{0} \quad (q \in F_6 \setminus A_Y).$$

Proposition 11 (Surjective backward map). *The projection $p_{\text{ud}} : \text{Grand} \rightarrow \text{Gen}$ is surjective:*

$$p_{\text{ud}} \circ z = \text{id}_{\text{Gen}}.$$

Proof. The zero extension preserves the u,d-face of m and resolves all other channels as inactive. Projecting back to the u,d-face therefore returns exactly m . $\square$

Surjectivity gives power: a Grand Meson can always be read as an ordinary General Meson. The same operation also carries responsibility, because the four hidden channels may contain active structure.

Definition 12 (Downgrade residue). For $g \in \text{Grand}$, define the downgrade residue

$$R_{\perp}(g) := g|_{F_6 \setminus A_Y}.$$

A downgrade $g \Downarrow m$ is physically admissible when

$$p_{\text{ud}}(g) = m \quad \text{and} \quad R_{\perp}(g) \in \text{Cl}(g \Downarrow m).$$

If $R_{\perp}(g) \neq \mathbf{0}$ and the residue is not included in the channel closure, the downgrade is inadmissible.

Remark 13 (Fusion and fission). Fusion is the admissible downgrade: the active extra axes are collapsed into the General Meson face only with an explicit residual channel. Fission is the inadmissible downgrade: the same projection is attempted while the extra axes are discarded as if they had never contributed. In the Grand Meson channel discipline, fission is not a physical mystery; it is an unclosed residual channel.

7 QCD Generalization

The Grand Meson generalizes the QCD mesonic flavor sector by refining each color-singlet quark–antiquark bilinear into a closed pair of flavor axes. Let the three flavor axes be

$$A_1 = \{u, d\}, \quad A_2 = \{c, s\}, \quad A_3 = \{t, b\}.$$

Definition 14 (Generation as internal direction). Let

$$G_3 = \{1, 2, 3\}, \quad H_2 = \{+, -\}.$$

Define

$$\Phi : G_3 \times H_2 \longrightarrow F_6$$

by

$$(1, +) \mapsto u, \quad (1, -) \mapsto d, \quad (2, +) \mapsto c, \quad (2, -) \mapsto s, \quad (3, +) \mapsto t, \quad (3, -) \mapsto b.$$

Then

$$A_i = \Phi(\{i\} \times H_2).$$

Proposition 15 (Three generations are Grand Meson directions). *The six-flavor Grand Meson decomposes as*

$$\text{Grand} = \text{Pol}^{F_6} \cong \text{Pol}^{G_3 \times H_2} \cong (\text{Pol}^{H_2})^{G_3}.$$

Thus the three observed quark generations are not external decorations of the six-channel configuration. They are the three internal directions of the Grand Meson, each carrying one up-type and one down-type face.

Definition 16 (Top-axis admissibility). The third direction

$$A_3 = \{t, b\}$$

is not an ordinary top-meson axis. It is a weak-virtual channel axis. For a channel $q\bar{r}$, define

$$\text{Top}(q\bar{r}) \iff (q = t) \text{ or } (r = t).$$

An ordinary mesonic readout is admissible only when $\text{Top}(q\bar{r})$ is false. If $\text{Top}(q\bar{r})$ is true, the channel must be read through a weak-virtual channel closure

$$\text{WCl}(q\bar{r}) \in \text{Cl}.$$

Proposition 17 (Unclosed top readout is GM-inadmissible). *Let $(i, j, O_{q\bar{r}})$ be a Grand Meson lifted channel. If $\text{Top}(q\bar{r})$ holds and $\text{WCl}(q\bar{r}) \notin \text{Cl}$, then the attempt to read $(i, j, O_{q\bar{r}})$ as an ordinary meson is inadmissible. Equivalently, the channel is Grand-inadmissible:*

$$\text{Top}(q\bar{r}) \wedge \text{WCl}(q\bar{r}) \notin \text{Cl} \implies \text{GM-inadmissible}.$$

Definition 18 (Grand Meson CKM lift). Let

$$U_L = (u_L, c_L, t_L)^T, \quad D_L = (d_L, s_L, b_L)^T,$$

and let $V_{\text{CKM}} = (V_{ij}) \in U(3)$ be the charged-current flavor-mixing matrix. The ordinary weak current is

$$J_+^\mu = \bar{U}_L \gamma^\mu V_{\text{CKM}} D_L = \sum_{i,j=1}^3 V_{ij} \bar{u}_{iL} \gamma^\mu d_{jL}.$$

The Grand Meson lift places each CKM coefficient on its axis pair:

$$\Gamma_{\text{GM}}^{\text{CKM}}(V_{ij}) := (i, j, V_{ij}).$$

Proposition 19 (CKM recovery). *Let $\pi_{\text{CKM}}(i, j, V_{ij}) = V_{ij}$. Then forgetting the Grand Meson axis labels recovers the ordinary CKM current:*

$$\sum_{i,j} \pi_{\text{CKM}}(\Gamma_{\text{GM}}^{\text{CKM}}(V_{ij})) \bar{u}_{iL} \gamma^\mu d_{jL} = J_+^\mu.$$

Thus CKM mixing is the charged-current axis-mixing layer of the Grand Meson, not a modification of QCD color dynamics.

The ordinary QCD mesonic operator basis is

$$\mathcal{Q}_{\text{mes}} := \left\{ O_{q\bar{r}} := \sum_{a=1}^3 q^a \bar{r}_a \mid q, r \in F_6 \right\}.$$

Standard channels are recovered inside this basis, for example

$$\pi^+ \sim O_{u\bar{d}}, \quad K^+ \sim O_{u\bar{s}}, \quad D^0 \sim O_{c\bar{u}}, \quad B^- \sim O_{b\bar{u}},$$

and

$$J/\psi \sim O_{c\bar{c}}, \quad \Upsilon \sim O_{b\bar{b}}, \quad \eta, \eta' \sim \text{span}\{O_{u\bar{u}}, O_{d\bar{d}}, O_{s\bar{s}}\}.$$

Thus cross-generation mesons such as $u\bar{s}$, $c\bar{d}$, $b\bar{u}$, and $c\bar{s}$ are ordinary members of the same basis. The Grand Meson lift is

$$\Gamma_{\text{GM}}(O_{q\bar{r}}) := (i, j, O_{q\bar{r}}) \quad \text{where } q \in A_i, \quad r \in A_j.$$

Thus

$$\mathcal{M}_{\text{GM}} := \{ (i, j, O_{q\bar{r}}) \mid q \in A_i, \quad r \in A_j, \quad 1 \leq i, j \leq 3 \}.$$

Proposition 20 (Conservative QCD generalization). *Let*

$$\pi_{\text{QCD}}(i, j, O_{q\bar{r}}) = O_{q\bar{r}}.$$

Then

$$\pi_{\text{QCD}} \circ \Gamma_{\text{GM}} = \text{id}_{\mathcal{Q}_{\text{mes}}}.$$

Therefore the Grand Meson is a conservative generalization of the QCD mesonic flavor sector: it adds the three-axis pair closure (i, j) while preserving the color-singlet QCD operator.

Definition 21 (Grand Meson source Lagrangian). Write the quark-sector QCD Lagrangian as

$$\mathcal{L}_{\text{QCD}} = -\frac{1}{4} G_{\mu\nu}^A G^{A\mu\nu} + \sum_{q \in F_6} \bar{q}(i\gamma^\mu D_\mu - m_q)q.$$

The Grand Meson source lift is

$$\mathcal{L}_{\text{GM}}[J] := \mathcal{L}_{\text{QCD}} + \sum_{i,j=1}^3 \sum_{\substack{q \in A_i \\ r \in A_j}} J_{ij}^{q\bar{r}} O_{q\bar{r}}.$$

Here $J_{ij}^{q\bar{r}}$ is not a new color force. It is the closed Grand Meson source attached to the axis pair (i, j) .

Proposition 22 (Lagrangian recovery). *Setting all Grand Meson sources to zero recovers QCD:*

$$\mathcal{L}_{\text{GM}}[0] = \mathcal{L}_{\text{QCD}}.$$

Forgetting the axis-pair labels sends the lifted source terms back to the ordinary mesonic source sector

$$\sum_{q,r \in F_6} J^{q\bar{r}} O_{q\bar{r}}.$$

Thus the Grand Meson generalizes the mesonic Lagrangian layer by adding axis-channel closure, not by altering the color-gauge core.

Definition 23 (Grand Meson quantum numbers). Let Q_q be the electric charge of quark q , and let $e_q \in \mathbb{Z}^{F_6}$ be the flavor unit at channel q . For a color-singlet mesonic channel with spin-parity label σ , define

$$\nu_{\text{QCD}}(q, \bar{r}; \sigma) := (\sigma, Q_q - Q_r, e_q - e_r, B = 0, \mathbf{1}_c).$$

The Grand Meson lift is

$$\nu_{\text{GM}}(q, \bar{r}; \sigma) := (i, j, \nu_{\text{QCD}}(q, \bar{r}; \sigma)), \quad q \in A_i, r \in A_j.$$

Here σ may be J^{PC} for neutral channels; for charged or open-flavor channels it is the corresponding spin-parity sector without an unavailable charge-conjugation label.

Proposition 24 (Quantum-number recovery). *If $\pi_\nu(i, j, \nu) = \nu$, then*

$$\pi_\nu \circ \nu_{\text{GM}} = \nu_{\text{QCD}}.$$

Thus the Grand Meson generalizes mesonic quantum numbers by adding only the axis-pair quantum number (i, j) , while preserving charge, flavor, baryon number, color singletness, and the spin-parity sector.

Definition 25 (Grand Meson observables). For QCD mesonic operators $O_{q\bar{r}}$, write the ordinary two-point observable as

$$C_{q\bar{r}, s\bar{t}}(x - y) := \langle O_{q\bar{r}}(x) O_{s\bar{t}}^\dagger(y) \rangle.$$

The Grand Meson lift is the axis-closed observable

$$C_{ij,kl}^{\text{GM}}(q\bar{r}, s\bar{t}; x - y) := (i, j, k, l, C_{q\bar{r}, s\bar{t}}(x - y)),$$

where $q \in A_i, r \in A_j, s \in A_k, t \in A_l$.

Proposition 26 (Observable recovery). *Let $\pi_{\text{obs}}(i, j, k, l, C) = C$. Then*

$$\pi_{\text{obs}}(C_{ij,kl}^{\text{GM}}(q\bar{r}, s\bar{t}; x - y)) = C_{q\bar{r}, s\bar{t}}(x - y).$$

Thus masses, widths, and spectral poles read from ordinary QCD correlators are preserved, while the Grand Meson retains which axis-pair channel made the observable readable.

Definition 27 (Grand Meson conservation law). Let $\mathbf{a}_{ij}$ be the unit basis element of $\mathbb{Z}^{3 \times 3}$. For $q \in A_i$ and $r \in A_j$, define

$$K_{\text{QCD}}(q\bar{r}) := (Q_q - Q_r, e_q - e_r, B = 0, \mathbf{1}_c)$$

and lift it to

$$K_{\text{GM}}(q\bar{r}) := (\mathbf{a}_{ij}, K_{\text{QCD}}(q\bar{r})).$$

Proposition 28 (Conservation-law recovery). *For an oriented mesonic process $\mathcal{P} = \sum_\alpha \eta_\alpha(q_\alpha \bar{r}_\alpha)$, ordinary QCD conservation is*

$$\sum_\alpha \eta_\alpha K_{\text{QCD}}(q_\alpha \bar{r}_\alpha) = 0.$$

The Grand Meson conservation condition is

$$\sum_\alpha \eta_\alpha K_{\text{GM}}(q_\alpha \bar{r}_\alpha) = (R_{\text{axis}}(\mathcal{P}), 0).$$

Thus QCD conservation is recovered by forgetting the axis residue R_{axis} . A closed Grand Meson process has $R_{\text{axis}} = 0$; otherwise the nonzero axis residue must be closed rather than ignored.

At this level QCD becomes concise: its mesonic flavor sector is the union of one-axis terms ($i = i$) and two-axis terms ($i \neq j$) among the three Grand Meson axes. The dynamics of color is not discarded; it is the singlet contraction already present in $O_{q\bar{r}}$. Channels touching t are retained as weak-virtual channels satisfying closure, not as ordinary top mesons.

8 Einstein's Gate

The Grand Meson must also avoid the classical aether failure. It is not a hidden medium with an absolute state of rest. It is a local six-channel mediator whose outputs must be readable without introducing an absolute frame. Einstein's Gate is not a claim of Einstein's endorsement. It names the admissibility condition inherited from relativity: no mesonic theory may reintroduce an absolute hidden medium.

Convention 29 (No absolute mesonic frame). No equation in the Grand Meson theory may depend on a privileged global rest frame of the mediator. Only resolved flavor channels, local channel closure, and observer-readable projections may be used. For a Lorentz transformation Λ , the flavor-axis labels (i, j) are internal labels and remain fixed, while spacetime arguments and tensor indices transform in the ordinary way:

$$(i, j, O_{q\bar{r}}(x)) \mapsto (i, j, O'_{q\bar{r}}(\Lambda x)).$$

Observable statements must therefore be Lorentz scalars or covariant tensor statements after forgetting the internal Grand Meson labels.

This convention is Einstein's Gate. It does not ban mediation. It bans absolute mediation. The Grand Meson is therefore not a return to the classical aether. It is a channel-resolved local mediator that must survive projection into admissible observations.

9 Why $n = 3$ Is the Admissible Case

Formally one may define an n -meson by

$$\text{Mes}_n := \text{Pol}^{2n}.$$

The case $n = 1$ is the General Meson. The case $n = 3$ is the Grand Meson.

The reason to stop at $n = 3$ is not that higher n is impossible, and it is not a proof of why nature has exactly three generations. It is that the known six-flavor quark sector already factors as

$$F_6 \cong G_3 \times H_2.$$

Therefore $n = 3$ is the exact internal-generation cover of the present mesonic flavor sector:

$$\text{Read}_{G_3}(\text{Mes}_3) = \text{Read}_{G_3}(\text{Pol}^{G_3 \times H_2}).$$

For $n > 3$, additional axes require additional readout structure. Such structure may be meaningful in a later theory, but it is not required for the present six-flavor sector:

$$\text{Read}_{G_3}(\text{Mes}_n) \not\supset \text{Read}_{G_3}(\text{Mes}_3) \quad (n > 3).$$

Thus $n = 3$ is the first complete and last admissible case for the present Grand Meson construction: complete because it covers the known three-generation flavor sector, admissible because the theory does not ask the reader to chase extra internal axes without new physical content.

10 Admissible Transition

Admissibility is not the absence of change. It is the channel closure of change. Let a transition be written

$$u : g \rightarrow g'.$$

The transition may change channels, support, or observable posture. It is inadmissible only when the change crosses a channel boundary without closure.

Definition 30 (Grand-admissible transition). Let $\text{Cl}(u)$ be the channel closure of a transition $u : g \rightarrow g'$. The transition is Grand-admissible when every changed channel is closed:

$$\{q \in F_6 \mid g(q) \neq g'(q)\} \subseteq \text{Cl}(u).$$

It is Grand-inadmissible when a changed channel is absent from $\text{Cl}(u)$.

Definition 31 (Physical admissibility). Grand-admissibility is channel admissibility, not full physical permission. A transition is physically admissible only when

$$\text{PhysAdm}(u) := \text{GrandAdm}(u) \wedge \text{Cons}_{\text{QCD}}(u) \wedge \text{Lag}_{\text{GM}}(u) \wedge \text{Obs}_{\text{GM}}(u).$$

Here Cons_{QCD} means that the conservation condition closes, Lag_{GM} means that the transition has a defined Lagrangian-source layer, and Obs_{GM} means that an observable readout exists.

Equivalently,

$$\text{inadmissible} = \text{unclosed discontinuity}.$$

The discontinuity itself is not the violation. The violation is that the boundary crossing occurred while the theory failed to enter it into the channel closure.

11 Conclusion

The Grand Meson is not proposed as an omnipotent meson. It is proposed as an admissible $n = 3$ general-purpose meson:

$$\text{Grand} = \text{Pol}^6.$$

It conservatively recovers the General Meson by a surjective projection, replaces missing channels with resolved flavor channels, avoids an absolute aether frame, and treats admissibility as channel closure rather than immobility. Its backward compatibility is therefore responsible: extra axes may be fused into the older face only with a residual channel closure, while unclosed fission is inadmissible.

The path is therefore narrow. Begin with the $u \leftrightarrow d$ eye that already works. Open $c \leftrightarrow s$ and $t \leftrightarrow b$ without breaking it. Stop at $n = 3$ until a future theory can make higher axes readable without turning them into a burden of infinite explanation.

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